

Event Sensitive Time-Mixture of Experts (EST-MoE): Enhancing Time Series Forecasting with External Factors

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Abstract

In this project, I propose the Event Sensitive Time-Mixture of Experts (EST-MoE), a novel extension of the traditional Time-MoE model, designed to enhance time series forecasting by incorporating a wide range of external event information. While Time-MoE efficiently handles time series data using a sparse Mixture of Experts (MoE) structure, EST-MoE integrates contextual encoding of external factors such as personal income, inflation, GDP growth, natural disasters, wars, exchange rate fluctuations, and social trends. This approach allows for more accurate and situation-aware predictions by dynamically selecting experts based on both temporal data and external contextual variables. The model employs a context-sensitive gating network and multi-resolution forecasting to handle both short-term and long-term predictions. Additionally, EST-MoE incorporates contextual sensitivity training to improve its adaptability and performance under varying external conditions. This work demonstrates that EST-MoE improves prediction accuracy and robustness compared to the traditional Time-MoE model, making it highly suitable for real-world applications where external events significantly influence outcomes.

1. Introduction

Time series forecasting is a crucial task in various domains, including finance, economics, and industry, where accurate predictions can significantly influence decision-making processes. Traditional time series models focus on capturing the temporal patterns within the data, but often overlook the impact of external events, such as economic changes or social disruptions, which can dramatically influence outcomes. The ability to incorporate these external factors is vital for improving the accuracy and robustness of predictions, especially in volatile environments.

With the advent of modern technology and the exponen-

tial increase in data sharing, the impact of external factors on forecasting has grown even more significant. Today, we live in an era where real-time information is continuously being produced and shared across social media platforms, digital networks, and various other channels. Events like economic shifts, natural disasters, political upheavals, and social trends can now spread faster and have a broader influence than ever before. As a result, the traditional models that focus solely on historical data are insufficient to predict future trends accurately.

Time-MoE (Time Mixture of Experts) is a powerful model designed to handle time series data by utilizing a sparse Mixture of Experts (MoE) architecture. The model efficiently processes large-scale temporal data, selectively

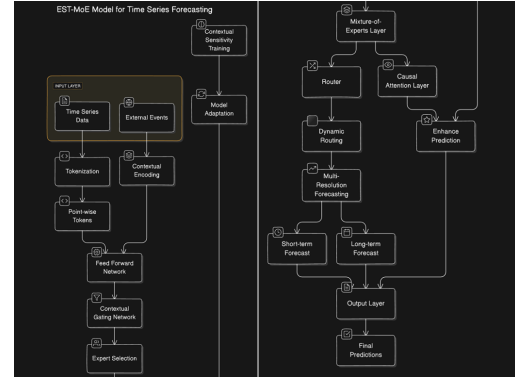


Figure 1. The Event Sensitive Time-Mixture of Experts (EST-MoE) model for time series forecasting. The diagram illustrates the overall architecture and flow of data through the model, starting with the input of Time Series Data and External Events. Key features include Contextual Encoding, and the Contextual Gating Network. The Mixture-of-Experts Layer further refines expert selection through Dynamic Routing. The Multi-Resolution Forecasting block generates both short-term and long-term forecasts simultaneously, while Contextual Sensitivity Training ensures that the model adapts its predictions based on external context changes, leading to enhanced prediction outcomes at the Output Layer.

activating only a subset of experts for each prediction. This sparse activation reduces computational costs while maintaining high prediction capacity. However, while Time-MoE excels at modeling time series data, it does not explicitly account for external event information, which is becoming increasingly important for precise forecasting in today's dynamic environments. Events like natural disasters, economic shifts, or societal trends can drastically alter the expected outcomes, and models that fail to incorporate these factors often produce inaccurate results.

To address this gap, I propose the Event Sensitive Time-Mixture of Experts (EST-MoE) model. This new model builds upon the Time-MoE framework by introducing external context sensitivity, enabling the model to dynamically adjust its predictions based on both temporal data and a variety of external factors such as personal income growth, inflation, GDP fluctuations, natural disasters, wars, exchange rate changes, and social trends. The inclusion of these factors allows the model to adapt to significant real-world events and produce more accurate, context-aware predictions.

At the core of EST-MoE are several key enhancements over the original Time-MoE:

- **Contextual Encoding:** EST-MoE incorporates additional input features representing external events. This external context is encoded alongside the time series data, allowing the model to consider both time-based patterns and the influence of external factors.
- **Contextual Gating Network:** A modified gating network selects the appropriate experts based not only on the time series data but also on the external context. This enables the model to activate experts that are most relevant to both the temporal data and the surrounding events.
- **Multi-resolution Forecasting:** EST-MoE provides short-term and long-term forecasting capabilities, allowing the model to predict across different time scales simultaneously. This feature is essential for handling both immediate responses to events and long-term trends.
- **Dynamic Routing:** The model dynamically routes inputs to different experts based on the combined influence of the temporal data and external context, ensuring that the most appropriate experts are utilized for each prediction.
- **Contextual Sensitivity Training:** The model is trained to be sensitive to external events, learning how changes in the external context should influence its predictions. This allows EST-MoE to adapt and improve its performance under various conditions.

Moreover, with the increasing scale of data sharing and the real-time availability of external event information, the influence of these factors on future trends has become even more profound. Modern technology has enabled vast amounts of information to be shared instantaneously across digital networks, meaning that events like global pandemics, political disruptions, or social movements can spread rapidly and influence markets and industries at an unprecedented rate. The ability to make multi-resolution predictions enables organizations to adjust their strategies for both immediate needs and long-term planning. This adaptability is crucial in sectors such as finance, where short-term volatility needs to be balanced with long-term investment goals, or in retail, where sudden changes in market conditions demand quick responses.

1.1. Proposed Solution: EST-MoE

The EST-MoE model is designed to improve the accuracy of time series forecasting by addressing the limitations of existing models that focus solely on temporal data. Through the incorporation of contextual encoding and the development of a contextual gating network, the model can dynamically adjust its predictions based on both temporal and external factors. Dynamic routing ensures that the appropriate experts are selected based on the specific conditions at hand, while multi-resolution forecasting allows the model to operate effectively across different time horizons. Finally, contextual sensitivity training enables the model to learn how external events should influence its predictions, leading to more accurate and reliable forecasts in the presence of external disruptions.

1.2. Key Contributions

If successful, this work will make the following major contributions:

1. **Enhancing time series forecasting with external event sensitivity:** EST-MoE introduces a novel approach by incorporating external factors such as economic conditions, social trends, and natural events into time series forecasting. This integration will result in more context-aware and accurate predictions, addressing a critical gap in existing models that focus solely on temporal data.
2. **Innovative contextual gating mechanism:** A new contextual gating network is developed to dynamically select the most relevant experts based on both temporal patterns and external contextual information. This mechanism enables more precise and adaptive expert selection, significantly improving model performance in diverse environments.
3. **Contextual sensitivity training for adaptability:** By incorporating contextual sensitivity training, EST-

MoE learns to adjust its predictions in response to changing external conditions, making it more robust in dynamic and unpredictable environments. This approach improves the model's adaptability to real-world scenarios where external events significantly impact outcomes.

In summary, EST-MoE offers a significant advancement in time series forecasting by addressing the critical need for context-sensitive predictions. This model has the potential to provide more accurate and reliable results in industries where external events have a significant impact on outcomes.

2. Background

Time series forecasting is fundamental in various domains such as finance, economics, and industry. Traditional models like ARIMA (AutoRegressive Integrated Moving Average) and LSTM (Long Short-Term Memory) have been pivotal in modeling time series data. ARIMA, a widely used statistical approach, models time series by explaining its autocorrelations, typically used for short-term forecasting. However, ARIMA models often struggle with nonlinear data and long-term forecasts [2] [1].

On the other hand, LSTM is a recurrent neural network variant specifically designed to handle long-range dependencies in sequential data through memory gates, making it suitable for capturing complex temporal patterns. LSTMs outperform ARIMA in most cases, especially for long-term forecasting, due to their capability to capture nonlinear relationships [1]. Despite their performance, LSTM models do not inherently account for external events, limiting their predictive power in dynamic environments influenced by exogenous factors like economic shocks or natural disasters.

2.1. Time-MoE and the Mixture of Experts Architecture

Recent advancements have introduced the Time-MoE model, designed to scale time series forecasting models through a sparse Mixture of Experts (MoE) architecture [3] [2]. The Time-MoE selectively activates a subset of experts, optimizing computational efficiency while maintaining high model capacity for large-scale forecasting tasks. Although Time-MoE demonstrates exceptional performance, it also falls short in integrating external events, making it less robust when handling real-world, event-sensitive scenarios [3].

2.2. Gaps in Current Solutions

While ARIMA and LSTM excel in different aspects of time series analysis, both have limitations in terms of sensitivity to external factors. Recent efforts, such as sparse MoE

models, have focused on scaling the model size and improving computational efficiency but still do not incorporate real-time external event sensitivity into their predictions. EST-MoE (Event Sensitive Time-Mixture of Experts) seeks to fill this gap by embedding contextual information like economic indicators, natural disasters, and societal trends, thus offering a more robust and accurate forecasting model in complex, real-world environments [2] [3].

By building on the Time-MoE architecture, EST-MoE enhances the ability to dynamically adjust predictions based on external factors, providing significant improvements in forecasting accuracy across various industries.

3. Method

In this section, I detail the methodology used to build the Event Sensitive Time-Mixture of Experts (EST-MoE) model. This model extends the existing Time-MoE framework by incorporating external event sensitivity into time series forecasting. The following subsections provide an overview of the key components of the model, the dataset used, and the approach to training and evaluation.

3.1. Model Architecture

The EST-MoE model builds upon the Mixture of Experts (MoE) architecture, utilizing a sparse activation mechanism to select only the most relevant experts for each input sequence. However, to enhance the model's performance in real-world scenarios, I introduce the following modifications:

- Contextual Encoding:** External event information such as personal income growth, inflation rates, GDP fluctuations, natural disasters, and wars is encoded as additional features alongside the time series data. These contextual features allow the model to account for external factors influencing the predictions.
- Contextual Gating Network:** The gating network is modified to consider both temporal data and contextual information when selecting experts. The gating network ensures that the appropriate experts are activated based on the influence of external events.
- Multi-Resolution Forecasting:** The model generates forecasts at both short-term and long-term intervals. This allows the EST-MoE to handle immediate responses to external events while also capturing long-term trends.
- Dynamic Routing:** The inputs are dynamically routed to different experts based on the combined influence of the temporal and contextual data. This mechanism ensures that the model adapts its predictions to the most relevant experts under varying external conditions.

5. **Contextual Sensitivity Training:** The model is trained to learn the sensitivity of the output to changes in the external context, ensuring that it adjusts predictions accordingly during significant external disruptions.

3.2. Math Behind the Model

The mathematical formulations behind each component of the EST-MoE model are described below.

3.2.1 Contextual Encoding

The time series data at time step t is denoted by $\mathbf{x}_t$, and the external event data is represented as $\mathbf{e}_t$. The contextual encoding combines these two inputs to create an enriched input vector:

$$\mathbf{z}_t = [\mathbf{x}_t, \mathbf{e}_t]$$

Here, $\mathbf{z}_t$ is a concatenation of the time series data $\mathbf{x}_t$ and the external event data $\mathbf{e}_t$, which is then passed into the model.

3.2.2 Contextual Gating Network

The gating network decides which experts to activate for each input based on both temporal and contextual information. The gating mechanism is described as:

$$\mathbf{g}_t = \text{softmax}(W_g \mathbf{z}_t + b_g)$$

where W_g is the weight matrix and b_g is the bias. The gating probabilities $\mathbf{g}_t$ determine the contribution of each expert. The softmax function ensures that the output is a probability distribution over the available experts.

3.2.3 Multi-Resolution Forecasting

The model produces forecasts for both short-term and long-term intervals. Let $\hat{y}_t^{(s)}$ represent the short-term prediction and $\hat{y}_t^{(l)}$ represent the long-term prediction at time t . The final objective combines both predictions, and the loss function is defined as:

$$\mathcal{L} = \alpha_s \mathcal{L}_{\text{short-term}} + \alpha_l \mathcal{L}_{\text{long-term}}$$

where $\mathcal{L}_{\text{short-term}}$ and $\mathcal{L}_{\text{long-term}}$ are the losses for the short-term and long-term predictions, respectively, and α_s and α_l are weighting factors to balance the contributions of each term.

3.2.4 Dynamic Routing

Dynamic routing ensures that inputs are forwarded to the most relevant experts based on the current temporal and

contextual data. For each input $\mathbf{z}_t$, the routing to the k -th expert is determined by:

$$\mathbf{h}_t^k = f_{\text{expert}_k}(\mathbf{z}_t)$$

where f_{expert_k} is the function of the k -th expert. The final output of the model is a weighted combination of the outputs from all active experts, given by:

$$\hat{y}_t = \sum_{k=1}^K g_t^k \mathbf{h}_t^k$$

Here, g_t^k is the gating weight from the softmax function, and $\mathbf{h}_t^k$ is the output of the k -th expert.

3.2.5 Contextual Sensitivity Training

To make the model sensitive to changes in external conditions, the training process incorporates a custom loss function that penalizes the model when it fails to react appropriately to significant changes in the external context. The loss function is augmented as follows:

$$\mathcal{L}_{\text{context}} = \mathcal{L}_{\text{original}} + \lambda \cdot \sum_t \|\Delta \mathbf{e}_t \cdot \Delta \hat{y}_t\|$$

where $\Delta \mathbf{e}_t$ is the change in external context at time t , $\Delta \hat{y}_t$ is the corresponding change in the prediction, and λ is a hyperparameter controlling the sensitivity of the model to external events. This term encourages the model to adjust its predictions in response to significant changes in the external environment.

3.3. Datasets

The model will be trained and evaluated on a mixture of both traditional time series datasets and datasets enriched with external event data. Specifically, the following datasets will be used:

- **M4 Dataset:** The M4 Dataset is a comprehensive dataset widely used in time series forecasting competitions. It consists of 100,000 time series data points collected across different domains, such as finance, industry, demographics, and macroeconomics. The dataset covers varying time frequencies (e.g., yearly, quarterly, monthly, weekly) and is designed to provide a robust basis for evaluating time series forecasting models. M4 is commonly used to benchmark forecasting models across multiple domains, making it an ideal substitute for the unavailable Time-300B dataset.
- **Federal Reserve Economic Data (FRED):** FRED is a comprehensive database maintained by the Federal Reserve Bank of St. Louis, providing economic data from various US and international sources. The dataset includes a wide range of economic indicators such as:

- GDP growth rates
- Unemployment rates
- Inflation and Consumer Price Index (CPI)
- Interest rates and monetary policy data
- Federal debt and deficit levels

FRED is crucial for macroeconomic forecasting and assessing economic trends over time.

- **World Bank Open Data:** The World Bank Open Data platform provides extensive global development indicators and economic statistics across numerous countries. The data available includes:

- GDP and Gross National Income (GNI)
- Employment and labor market indicators
- Trade and investment data
- Health and education statistics
- Poverty and inequality measures

These indicators are essential for evaluating economic development patterns and performing global economic analysis in a time series context.

- **The International Disaster Database (EM-DAT):** The EM-DAT database, maintained by the Centre for Research on the Epidemiology of Disasters (CRED), offers detailed data on natural and technological disasters. Data available includes:

- Natural disasters such as earthquakes, floods, hurricanes, and droughts
- Data on the human and economic impact of these disasters (e.g., casualties, displacement, and damage)
- Global distribution of disaster events by frequency and severity

This data allows models to incorporate the impact of natural disasters on economic and social patterns, improving forecasting accuracy in real-world event-sensitive scenarios.

3.4. Computational Resources

In this project, I plan to utilize both the BlueHive high-performance cluster and Google Colab's NVIDIA A100 GPU for training the EST-MoE model. The BlueHive cluster will be used for parallel processing of large-scale datasets, while the A100 GPU will be used for fine-tuning the model and running experiments that require fast training cycles. Together, these resources will provide sufficient computational capacity to handle the complexity of

the model and the size of the datasets. Distributed training on BlueHive, combined with the GPU's capability in Colab, will ensure the EST-MoE model is trained efficiently and within a reasonable time frame.

3.5. Evaluation

For the evaluation, I will measure the performance and success of the Event Sensitive Time-Mixture of Experts (EST-MoE) model by comparing it against other standard models such as Time-MoE and traditional time series models like ARIMA and LSTM. The following evaluation methods will be employed:

1. Comparison with Baseline Models:

- **Time-MoE:** The original Time-MoE model will serve as a baseline for comparison. EST-MoE's enhanced ability to account for external events will be evaluated to demonstrate its improvements over the standard Time-MoE.
- **ARIMA and LSTM:** Traditional time series forecasting models will also be used to benchmark the performance of EST-MoE. By comparing the results, I will highlight how EST-MoE outperforms models that do not incorporate external context.

2. Metrics for Evaluation:

- **Mean Squared Error (MSE):** For continuous value predictions, MSE will be the primary metric to assess how well the model predicts numerical outputs.
- **Accuracy:** For classification tasks such as predicting event categories, accuracy will measure the percentage of correct predictions made by the model.
- **F1-Score:** This will be used to evaluate the robustness of the model in handling imbalanced data, particularly for event classification tasks.
- **R-Squared (R^2):** This will measure the proportion of variance explained by the model for regression tasks, indicating how well the model fits the data.

3. Computational Efficiency:

- **Training Time:** The time required to train EST-MoE compared to Time-MoE and traditional models will be recorded and analyzed to evaluate the computational cost.
- **Memory Usage:** Memory consumption during training and inference will be measured to determine the efficiency of the model given its increased complexity.

4. Timeline

Milestone	Projected Date
Dataset Preparation	10/25/2024
Model Implementation	11/01/2024
Preliminary Testing	11/10/2024
Contextual Sensitivity Training	11/20/2024
Final Model Tuning	12/01/2024
Final Report Submission	12/05/2024

Table 1. Project timeline with key milestones and deadlines

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